Michael Chen

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Experience

AI Engineer October 2025 – Present
Epoch (Zerg AI) *Remote*

- Built agent-runtime features for ZTC, Zerg's terminal coding agent (TypeScript): parallel tool dispatch, background sub-agents, worktree-isolated batch edits, and multi-agent workflows; cut resumed-session context 92%
- Engineered CI for Zergstack, Zerg's internal Render-style deploy platform on Fly.io, running GitHub Actions jobs on per-repo Sprite pools restored to clean checkpoints at about 1/4 the per-job cost of GitHub-hosted runners
- Designed a domain-agnostic multi-agent simulation kernel (typed LLM boundaries, agent memory, causal world graph, deterministic replay) that powers 14 corporate-crisis scenarios and 9 interactive worlds, with 1,847 tests
- Shipped two-way Slack sync and the agent runtime for ZergChat, a team chat where AI agents are members (Nuxt, PostgreSQL, Redis, Tauri); agents answered 56/56 probe questions over a 20K-message corpus
- As Forward Deployed Engineer for an aerospace company, built a continual-validation CI gate that scores each PR against a deterministic simulator, and an LLM + OCR invoice pipeline with zero regressions on 22,643 documents
- Scaled the customer's operations platform (Nuxt, PostgreSQL, Kubernetes) to 1,000 concurrent users at 100ms p95, fixed 190+ defects including SSO account takeover, and built a live MRP engine over ERP and PLM data

Software Engineer (Part-Time) June 2023 – Present
MOBIVOLT LLC *Livermore, CA*

- Created an AI-agent circuit design pipeline (KiCad, ngspice, FreeCAD) that turned a 6-line spec into a fabrication-ready voltage monitor for 20–80-cell stacks, with zero ERC/DRC errors and 36/36 simulation corners passing
- Built 10+ automated test systems streaming 100+ sensor channels for a $5M+ DOD fuel-cell program, then merged 9 of their GUIs into one cross-platform C#/Avalonia app that passed an 18M-frame soak with zero drops
- Migrated two fuel-cell controllers to MPLAB Harmony v3 on PIC32MZ (50 modules, 23K lines of C) with 3,900+ lines of C tests, and built hardware-in-the-loop simulators that run the firmware against plant models

Founder and President October 2023 – September 2025
Artificial Intelligence at UCR (AIR) *Riverside, CA*

- Founded UCR's first official AI organization, grew it to 50+ active members in 6 months, and raised workshop attendance 45%
- Taught 10+ technical workshops on PyTorch, machine learning, and LLM APIs, and published the curriculum on GitHub

Projects

Ornith-1.5 35B: Compressed MoE LLM for Apple Silicon August 2026

- Compressed a 35B mixture-of-experts LLM from 69 GB to 14 GB with router-weighted expert pruning (256 to 192 experts), mxfp4 quantization, and fewer active experts, so it runs on a 24 GB Mac mini
- Reached 49.8 tokens/s with 98.1% on a 42-task coding, math, and tool-call suite, beating an unpruned 3-bit build (36.7 tokens/s, 95.4%); 430+ downloads on Hugging Face

Kestrel: Distilled Text-to-Speech for Apple Silicon July 2026 – August 2026

- Distilled Kokoro-82M into a 10M-active-parameter MLX engine that renders a 163-second chapter in 0.27s on an M2 MacBook, 53× faster than stock, at parity ASR word error rate; released open weights on Hugging Face
- Found 5 divergences between MLX and PyTorch decoders by diffing tensors layer by layer; merged upstream fix to mlx-audio restored a 2.5 dB level loss and cut mel-cepstral distortion from 7.29 to 4.09 dB

PHiLIP: Personalized Human in Loop Image Production November 2023 – July 2024

- Won the AMD University Program Award ($2,500) among 8,200+ entrants at the AMD Pervasive AI Developer Contest; demoed live at AMD Advancing AI 2024
- Developed image generation with 2-second inference, 71 style presets, and fine-tuned PixArt-alpha on AMD ROCm

Education

University of California, Riverside Riverside, CA
B.S. in Computer Science, Minor in Entrepreneurship and Strategy; GPA 3.66/4.0, Dean's Honor List *June 2026*

Technical Skills

AI/ML: LLM Agents, MCP, Multi-Agent Systems, LLM Evaluation, RAG, Quantization, Distillation, vLLM, MLX, PyTorch, Hugging Face
Languages: Python, TypeScript, JavaScript, C, C++, C#, Go, Rust, SQL, Bash
Frameworks: Node.js, Vue/Nuxt, React/Next.js, FastAPI, .NET/Avalonia, PyQt, Tauri, Playwright
Data/Cloud: PostgreSQL, Redis, Supabase, Docker, Kubernetes, Helm, AWS, Fly.io, Modal, Vercel, GitHub Actions
Hardware: PIC32MZ, MPLAB Harmony v3, Arduino, KiCad, ngspice, FreeCAD, Hardware-in-the-Loop Testing